THE FIBRE-DIMENSION FUNCTION OF A PROPER MAP OF SPACES IS UPPER SEMI-CONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7226-7233 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7235-7244, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-base-change-dimension-fibres; Morphisms of Spaces, lemma openness-bounded-dimension-fibres.

Source reference. Original lemma 05F8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7183–7196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7244–7244. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S. Let $n_{X/Y}$ be the function on Y giving the dimension of fibres of f introduced in Lemma 05F8. Then $n_{X/Y}$ is upper semi-continuous.*

Proof. Let $Z_d = \{x \in |X| : \text{the fibre of } f \text{ at } x \text{ has dimension } > d\}$. Then Z_d is a closed subset of $|X|$ by Morphisms of Spaces, Lemma 04NT. Since f is proper $f(Z_d)$ is closed in $|Y|$. Since $y \in f(Z_d) \Leftrightarrow n_{X/Y}(y) > d$ we see that the lemma is true. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.